

SAYAN DAS

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Education

Jadavpur University

Aug 2023 - Jul 2027 (expected)

Bachelor of Science (Honours) in Mathematics with Research

Minors in Statistics and Computer Science

Maths courses: Real Analysis, Linear Algebra, Abstract Algebra, Ring Theory, Numerical Analysis, Metric Space & Differential Geometry, Ordinary Differential Equations.

NPTEL courses: [Algebraic Number Theory](#).

CS courses: Programming Languages, Database Management Systems, Data Structures & Algorithms in Python.

Stats courses: Probability & Descriptive Statistics, Probability Distributions and χ^2 -test.

Research Interests

I am primarily interested in number theory - both algebraic and analytic. In particular, I am interested in automorphic forms, subconvexity of L -functions, equidistribution, sieve theory, prime numbers, Iwasawa theory, and harmonic analysis on number fields i.e. Iwasawa-Tate theory.

Research Experience

Summer Internship in Analytic Number Theory

1 Jul 2025 - 31 Jul 2025

Supervisor: [Prof. Satadal Ganguly](#)

Indian Statistical Institute Kolkata

- Worked through Apostol's *Introduction to Analytic Number Theory*.
- Learnt about the distribution of primes, asymptotic estimates of arithmetic functions.
- Learnt about Dirichlet characters and L -functions in connection with Dirichlet's Theorem on primes in arithmetic progression
- Learnt about Fourier series for periodic arithmetic functions, Ramanujan sums and Gauss sums in connection with quadratic reciprocity.
- Culminated with an exposition of Selberg's proof of the Prime Number Theorem. [1]

Volunteering

Content Team Assistant Convenor

Sep 2024 - Present

[Jadavpur Math Society](#)

- Contributed to the Jadavpur Math Society magazine [2] and daily problems list.
- Set the question paper for and helped organise Mathemagician at JU's annual tech-fest Srijan 2025, which included checking and grading answer scripts.

Articles

[1] S. Das, *Selberg's elementary proof of the prime number theorem*, [Link to report](#), 2025.

[2] S. Das, *Fermat's last theorem for regular primes*, [Link to draft](#), 2025.

Projects

Numerical Analysis

Mar 2025 - May 2025

- Implementations of some common numerical algorithms in MATLAB.
- Contains algorithms for linear and nonlinear systems, eigenvalues, differential and integral equations.

Achievements

Algebraic Number Theory

May 2025

NPTEL, Indian Institute of Science Bangalore

- Achieved the highest score: **71%**.
- Distinguished myself as the only course topper per course **statistics**.

Jadavpur University BSc Math Entrance Exam

Jun 2023

- **Ranked 49** in the Other Boards General category.

Workshops

Winter School on CS Theory

Dec 2024

Indian Institute of Science Bangalore

- **One of only 3 students selected** to attend the winter school from Jadavpur University.
- Attended lectures on matching theory and differential privacy.

Skills

Programming Languages

C, C++, Python, SQL, MATLAB, C#, Java, JavaScript, HTML/CSS

Developer Tools

Make, CMake, Vim, Neovim, Git, PowerBI, VS Code, Visual Studio, IntelliJ, Eclipse, Unity, Unreal

APIs, Frameworks, & Libraries:

Pandas, NumPy, Matplotlib, PyTorch, Tensorflow, Scikit, Boost, SFML, Vulkan

References

1. Prof. Shamik Ghosh,
Professor, Department of Mathematics, Jadavpur University
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E-Mail: shamik.ghosh@jadavpuruniversity.in.
2. Prof. Subhas Chandra Mandal,
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